Candidate's name.....

School.....

P525/1
CHEMISTRY
Paper
July 2026
2¾ Hours

SCIENCE EDUCATORS' EXAMINATION BOARD

Uganda Advanced Certificate of Education

CHEMISTRY (Theory)

Paper 1

INSTRUCTIONS TO CANDIDATES

*This examination paper consists of **two sections**, **A** and **B**, comprising **six examination items**.*

- *Section A contains **two compulsory items**.*
- *Section B is divided into **Parts I and II**. Answer **one item from each part**.*
- *Answer **four items in total**.*
- *Any additional item(s) attempted **will not be marked**.*
- *All responses **must be written in the answer booklet(s) provided**.*
- *Begin **each item on a new page**.*

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SECTION A

Respond to **both** items from this section.

1. Lead-acid accumulators are widely used in automobiles and emergency power supplies because they are rechargeable and relatively inexpensive.
A battery manufacturing company plans to adopt a lead-acid accumulator to reduce production costs. Before making this decision, management requires a technical evaluation of the proposed battery.

The management is concerned about its operation and advantages, kinetic performance under different conditions, ability to supply at least 250 kJ mol^{-1} of electrical energy under standard conditions, thermodynamic feasibility of lead (II) sulphate formation, and suitability for large-scale production.

A student with knowledge in electrochemistry and thermodynamics was contacted to assist in addressing these concerns.

The following information is provided.

Electrode reactions

$$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^- \quad E° = +0.13\,V$$
$$PbSO_4(s) + 2H_2O(l) \rightarrow PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \quad E° = +1.46\,V$$

The company seeks to understand how these reactions affect battery operation and performance.

Table 1: Standard enthalpy changes

Reaction	ΔH° / kJ mol^{-1}
$H_2(g) + S(s) + 2O_2(g) \rightarrow H_2SO_4(l)$	–814.8
$H_2SO_4(l) \rightarrow H_2SO_4(aq)$	–91.4
$PbSO_4(s) \rightarrow PbSO_4(aq)$	–36.2
$Pb(s) + H_2SO_4(aq) \rightarrow PbSO_4(aq) + H_2(g)$	–120.5

The company is unsure about the thermodynamic feasibility of lead (II) sulphate formation and its impact on battery operation.

Table 2: Kinetic data

Condition	Initial current / A	Time to reach steady voltage / s
298 K	0.50	110
318 K	0.72	65

The company is interested in how operating conditions affect reaction rate and efficiency.

Task

Prepare a report addressing the management's concerns using the information provided.

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2. Volatile liquid mixtures and weak acid systems are essential in solvent recovery and industrial cleaning.

A chemical manufacturing company in Uganda uses a solvent mixture containing methanol and cyclohexane. After production, the mixture is separated by simple distillation so that the solvents can be recovered and reused. However, the recovered solvents do not attain the required purity, leading to solvent losses.

To investigate the problem, the quality control laboratory measured the boiling points and methanol compositions in the liquid and vapour phases.

The results are shown in **Table 2**.

Table 2

Boiling point of mixture (°C)	Mole fraction of methanol in vapour	Mole fraction of methanol in liquid
70	0.12	0.27
60	0.13	0.47
55	0.50	0.56
57	0.82	0.69
61	0.94	0.83

The boiling points of pure cyclohexane and methanol are 81°C and 65°C, respectively.

The company also manufactures a cleaning formulation containing methanoic acid. During quality control, 50.0 cm^3 of 0.050 mol dm^{-3} methanoic acid was mixed with 50.0 cm^3 of 0.020 mol dm^{-3} sodium hydroxide at 25°C. During storage, 5.0 cm^3 of 0.020 mol dm^{-3} hydrochloric acid was accidentally added. The pKa of methanoic acid is 3.75. The company is investigating whether the formulation can maintain a stable pH and whether similar buffer systems could have other applications.

You were requested to investigate the problem and assist the company in understanding the causes of the solvent losses and the suitability of the cleaning formulation.

Tasks

(a) Analyse the quality control laboratory data to explain why solvent losses occur.
(b) Evaluate the suitability of the methanoic acid formulation for maintaining a stable pH and explain other applications of similar systems.

SECTION B
Part I

Respond to any **one item** in this part.

3. Chlorine and its compounds are widely used in water treatment because of their ability to destroy disease-causing microorganisms. Your class visited a district water treatment plant to study the role of Group VII elements in water purification and sanitation.

At the treatment plant, the process engineer provided the following data.

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Table 3

Halogen	Fluorine	Chlorine	Bromine	Iodine
Electronegativity	3.98	3.16	2.96	2.66
Boiling point (°C)	–188	–34	59	184

The engineer explained that the electronegativity and boiling point data could be used to compare, respectively, the reactivities and physical states of the halogens.

He further explained that chlorine reacts with water to produce substances responsible for disinfection, while chlorine and bromine react differently with sodium hydroxide depending on the reaction conditions.

He also emphasized that although chlorine is an effective disinfectant, its handling and use require strict safety measures because of its effects on human health.

You found the engineer's explanations convincing and decided to support them with scientific evidence.

Task

Prepare a presentation, based on the information obtained during the study tour, to justify the explanations provided by the process engineer.

4. The chemistry of Group II elements has many industrial applications, including the manufacture of lightweight alloys and materials for water storage.
A company that manufactures bicycle parts and drinking-water storage containers plans to use Group II elements in its production. The production team must select the most suitable Group II metal as a component of a low-melting-point alloy for joining bicycle parts and another for manufacturing drinking-water storage containers that will not react with the stored water.

During preliminary trials, they observed that the Group II metals reacted differently with water, and dilute sulphuric acid. They also found that beryllium behaved differently from the other Group II elements and exhibited properties similar to those of aluminium, and wondered how and why this was so, based on the reactions observed.

To support the selection process, a chemical engineer provided the following data.

Table 4

Element	Beryllium	Magnesium	Calcium	Strontium	Barium
Atomic radius (nm)	0.112	0.160	0.197	0.215	0.222
Melting point (°C)	1283	650	848	770	710

The company has invited schools to participate in an industrial problem-solving exercise, and you have been selected to represent your school.

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Task

(a) Using the information provided, select the most suitable Group II metal for each application based on the trends in the physical property data in the table 1

(b) Address the company's concerns

(c) Explain some environmental implications associated with the selected metal in part (a).

Part II

Respond to any **one item** in this part.

5. The growing demand for specialty organic chemicals and polymers has encouraged manufacturers to improve production efficiency and sustainability.
An organic chemistry industry in Mbale produces cosmetics and packaging materials. The company requires benzaldehyde and styrene, but its current stocks are exhausted.

Fig.1: Structure of Benzaldehyde

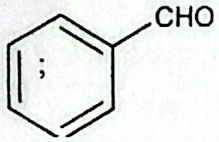


Fig.2: Structure of styrene

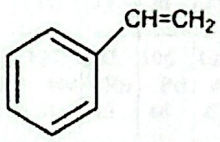
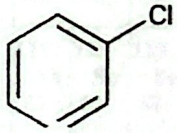


Fig.3. Structure of chlorobenzene



The available starting materials are ethanoic acid and chlorobenzene, which can be used to prepare benzaldehyde and styrene, respectively.

During pilot production, engineers found that residues of benzaldehyde and ethanoic acid could pose environmental risks if discharged untreated. An industrial chemist recommended treating the benzaldehyde residues with sodium hydrogensulphite and the ethanoic acid residues with warm acidified ethanol before disposal.

The company also requires guidance on the health implications of the compounds involved, why benzaldehyde suits its role.

Task

Prepare a scientific report advising the organic chemistry industry on how to address these concerns.

6. The chemical industry relies on organic compounds to manufacture a wide range of industrial products.
An organic chemistry laboratory received samples from a businessman who manufactures phenol-based mouthwash and alcohol-based hand sanitizers using propan-2-ol. During waste disposal, a cleaner accidentally mixed phenol with alkaline waste containing a chloride of ethanoic acid.

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In a separate incident, a batch of propan-2-ol was stored in a container that had previously contained hot concentrated orthophosphoric acid. The laboratory manager requested an investigation into the incidents.

Meanwhile, researchers investigated the use of 2-bromopropane as a starting material for preparing the samples received from the businessman in case they ran out of stock. The laboratory manager also sought advice on the properties, applications, and environmental impacts of the compounds involved.

Tasks

(a) Illustrate the reaction pathways for preparing the samples in case they run out of stock.
(b) Evaluate the physical properties that make the received samples suitable for their respective industrial applications and explain one other industrial application of each sample.
(c) Explain the chemical reactions involved in the laboratory incidents under investigation and advise the laboratory manager on **one** environmental impact of the major product produced in each incident.

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END

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Tl 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

		139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
		227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

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